

Surrogate-Assisted Diversity Estimation in Neural Ensemble Search

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Motivation and Approach

Problem

Develop an efficient method for Neural Ensemble Search (NES)

Challenge

Neural architecture search is computationally expensive, and constructing optimal ensembles leads to a exponential explosion of candidate ensemble configurations.

Proposed Approach

Employ a surrogate-assisted method by embedding architectures into a latent space, where similar networks are close and dissimilar ones are far.

Motivation

Efficient estimation of ensemble diversity enables better ensemble selection without exhaustive training.

Problem Statement

Let $\alpha \in \mathcal{A}$ denote a network architecture, and $\omega(\alpha)$ its trained weights. The output of a network on input $\mathbf{x}$ is $f_{\alpha}(\mathbf{x}, \omega(\alpha))$. Given a subset $S \subset \mathcal{A}$, the NES problem is formally defined as:

$$\begin{aligned} \min_S \mathcal{L}_{val} \left(\frac{1}{|S|} \sum_{\alpha \in S} f_{\alpha}(\mathbf{x}, \omega^*(\alpha)) \right) \\ \text{s.t. } \forall \alpha \in S : \omega^*(\alpha) = \arg \min_{\omega(\alpha)} \mathcal{L}_{train}(f_{\alpha}(\mathbf{x}, \omega(\alpha))) \end{aligned}$$

Challenges:

- ▶ Exponentially growing number of architectures
- ▶ Exponentially growing number of ensemble combinations

Surrogate Functions

Overview

We propose two surrogate functions:

- $f_{\text{acc}}^{\theta} : \mathcal{A} \rightarrow \mathbb{R}$ — predicts network accuracy. Trained by minimizing mean squared error (MSE):

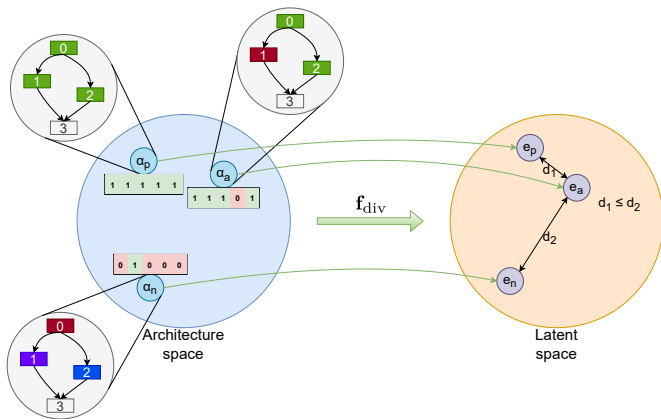
$$\theta^* = \arg \min_{\theta} \sum_i (f_{\text{acc}}^{\theta}(\alpha_i) - y_i)^2$$

- $\mathbf{f}_{\text{div}}^{\theta} : \mathcal{A} \rightarrow \mathbb{R}^d$ — maps architectures to a latent space. Trained with triplet loss to preserve diversity:

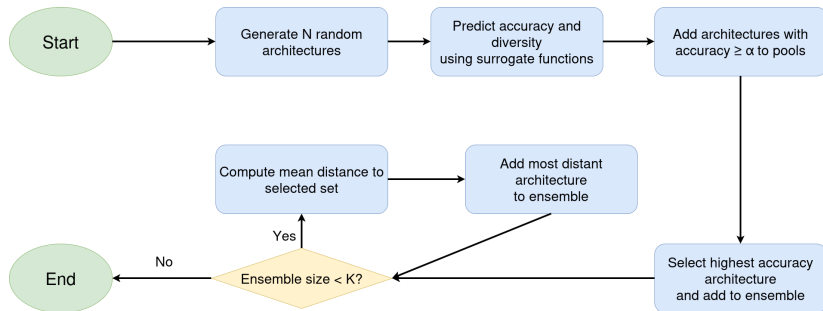
$$\theta^* = \arg \min_{\theta} \sum_{(\alpha_a, \alpha_p, \alpha_n)} \max \left(\|\mathbf{f}_{\text{div}}^{\theta}(\alpha_a) - \mathbf{f}_{\text{div}}^{\theta}(\alpha_p)\|_2^2 - \|\mathbf{f}_{\text{div}}^{\theta}(\alpha_a) - \mathbf{f}_{\text{div}}^{\theta}(\alpha_n)\|_2^2 + m, 0 \right)$$

where $(\alpha_a, \alpha_p, \alpha_n)$ denote anchor, positive, and negative architectures, $m \in \mathbb{R}$ denote margin.

Surrogate Diversity Function



Surrogate Assisted Ensemble Construction



The proposed ensemble construction algorithm.

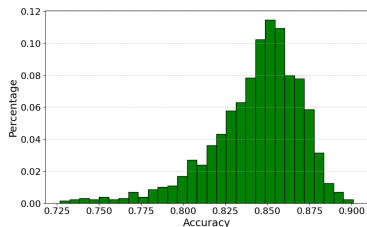
Goals of the Computational Experiment

On datasets FashionMNIST, CIFAR10, CIFAR100:

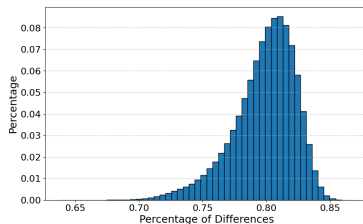
- ▶ Construct a training dataset of architectures and their evaluations
- ▶ Evaluate the surrogate models' ability to predict accuracy and diversity
- ▶ Assess the effectiveness of the ensemble construction strategy
- ▶ Compare surrogate-assisted method with DeepEns and RandomSearch

Dataset of Architectures

For each dataset, 3,000 neural architectures were partially trained. The train/validation split was set to 0.2/0.8 for all datasets. As an example, the resulting distributions for CIFAR10 are shown below:



Distribution of model accuracies



Distribution of model diversity

Component Ablation Study (CIFAR-100)

Contribution of each surrogate function

Method	Top-1 Acc. (%)	Avg. Model Acc. (%)
Random	84.41 $\pm$ 0.09	79.57 $\pm$ 0.19
Accuracy only	85.07 $\pm$ 0.10	80.54$\pm$0.12
Diversity only	84.77 $\pm$ 0.07	80.08 $\pm$ 0.21
Accuracy + Diversity	85.17$\pm$0.16	80.50 $\pm$ 0.12

- ▶ Joint optimization of accuracy and diversity consistently outperforms single-objective selection
- ▶ Explicit diversity modeling improves ensemble performance on complex tasks
- ▶ Random search yields higher disagreement but weaker individual models

Surrogate Performance vs Training Set Size

Type	Train Size	Spearman	R@10	R@50	R@100
Accuracy	0	0.017 $\pm$ 0.051	0.010 $\pm$ 0.030	0.076 $\pm$ 0.017	0.167 $\pm$ 0.021
	250	0.106 $\pm$ 0.106	0.040 $\pm$ 0.049	0.130 $\pm$ 0.039	0.228 $\pm$ 0.052
	500	0.220 $\pm$ 0.058	0.090 $\pm$ 0.083	0.196 $\pm$ 0.064	0.291 $\pm$ 0.034
	750	0.218 $\pm$ 0.079	0.120 $\pm$ 0.125	0.204 $\pm$ 0.061	0.291 $\pm$ 0.046
	1000	0.368 $\pm$ 0.050	0.120 $\pm$ 0.087	0.264 $\pm$ 0.073	0.330 $\pm$ 0.025
	1500	0.535 $\pm$ 0.060	0.170 $\pm$ 0.135	0.294 $\pm$ 0.073	0.366 $\pm$ 0.032
	2000	0.675$\pm$0.021	0.200$\pm$0.089	0.324$\pm$0.032	0.430$\pm$0.031
Diversity	0	-0.001 $\pm$ 0.029	0.019 $\pm$ 0.003	0.092 $\pm$ 0.005	0.179 $\pm$ 0.006
	250	-0.490 $\pm$ 0.042	0.029 $\pm$ 0.004	0.131 $\pm$ 0.008	0.249 $\pm$ 0.011
	500	-0.545 $\pm$ 0.035	0.031 $\pm$ 0.003	0.136 $\pm$ 0.005	0.261 $\pm$ 0.008
	750	-0.574 $\pm$ 0.025	0.029 $\pm$ 0.003	0.135 $\pm$ 0.006	0.259 $\pm$ 0.010
	1000	-0.602 $\pm$ 0.025	0.031 $\pm$ 0.003	0.142 $\pm$ 0.007	0.271 $\pm$ 0.012
	1500	-0.639$\pm$0.022	0.032$\pm$0.003	0.147$\pm$0.005	0.281$\pm$0.007
	2000	-0.637 $\pm$ 0.025	0.032$\pm$0.004	0.145 $\pm$ 0.007	0.278 $\pm$ 0.008

- **Accuracy surrogate:** Spearman correlation and Recall steadily improve with dataset size.
- **Diversity surrogate:** rapidly converges at $\sim$ 500–1000 samples; further increase yields diminishing returns.

Comparison of Surrogate-Assisted NES and DeepEns

Metric	Surrogate Ens.	DeepEns	Random
FashionMNIST			
<i>Ensemble size = 3</i>			
Top-1 Accuracy	95.3%	95.02%	95.01%
Average Model Accuracy	94.7%	94.55%	94.32%
NLL	0.263	0.266	0.265
CIFAR10			
<i>Ensemble size = 5</i>			
Top-1 Accuracy	97.80%	97.64%	97.56%
Average Model Accuracy	96.73%	96.71%	96.48%
NLL	0.208	0.210	0.212
CIFAR100			
<i>Ensemble size = 5</i>			
Top-1 Accuracy	85.17%	85.10%	84.41%
Average Model Accuracy	80.50%	80.97%	79.57%
NLL	0.692	0.682	0.718

Conclusion

- ▶ We propose a surrogate function for predicting architectural diversity.
- ▶ We introduce a surrogate-assisted framework for neural ensemble construction.
- ▶ Our method matches or outperforms DeepEns and Random Search on FashionMNIST, CIFAR-10, and CIFAR-100 by explicitly accounting for architectural diversity during ensemble construction.
- ▶ Ablation studies confirm that joint accuracy-diversity optimization is essential for complex tasks.
- ▶ The diversity surrogate converges rapidly (~ 500 – 1000 samples), making the approach computationally efficient.